

E-Commerce Sales Analysis

Findings Report

Dataset and Cleaning

The dataset covers UK-based online retail transactions from December 2010 through December 2011, totaling 536,641 rows. About 24.9% of rows had no CustomerID (likely guest checkouts) and were set aside for customer-level analysis. 5,268 exact duplicate rows were removed, and 1.7% of transactions were cancellations (InvoiceNo starting with "C").

Revenue and Losses

The largest single order (Invoice 581483) totaled \$168,469.60, but this exact order was fully cancelled and refunded in the same billing cycle, making it a wash rather than real revenue.

The top loss driver by far is "AMAZON FEE" at negative \$235,281.59, followed by a "MANUAL" adjustment line (negative \$146,784.46) and the cancelled bulk order above. These are largely bookkeeping and fee entries rather than typical product returns.

The United Kingdom accounts for the largest share of both total losses and sales volume, which is unsurprising since it is the dominant market. Return rate by country is a more useful lens than raw loss totals for identifying problem markets.

Time Patterns

Sales were broken out by month and day of week. Both show clear variation across the year and across the week, worth highlighting the specific peak periods once reviewed directly against the underlying charts.

Product-Level Returns

Products were ranked by return rate (returns divided by sales), surfacing which SKUs are disproportionately refunded. This is useful for flagging problem inventory, such as defective items or inaccurate listings, separately from high-volume expected returns.

Customer Segmentation (RFM)

Using Recency, Frequency, and Monetary scoring across roughly 4,338 customers with valid IDs, four segments emerged:

Segment	# Customers	Total Revenue
Champion	1,267	\$6,827,235.77
Loyal	845	\$1,051,599.98

At Risk	935	\$650,529.15
Lost	1,291	\$357,843.99

Key insight: Champions make up only 29% of customers but generate approximately 76% of total revenue. Lost customers form the largest single segment by headcount (1,291) but contribute the least value. Retention efforts are far better spent protecting and rewarding the Champion tier than trying to win back the Lost segment.